

Indian Institute of Information Technology Kottayam

Department of Computer Science and Engineering

CSE312: Software Architecture — Principles and Practices

User Interface Design and Implementation

LexFlow (“nyaya”)

AI-Powered Legal Case Assistant & Case Management System

Submitted By

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1. Introduction

This document is the UI Design and Implementation submission for LexFlow, built following the requirements, Use Case Diagram, and UML diagrams (Class, Sequence, Activity and State) submitted earlier for CSE312. The implemented product is client-branded as “nyaya” — the working title LexFlow used in the system-design documentation and the shipped brand name nyaya refer to the same system. The interface follows a dark, editorial visual language (navy background, gold accent, serif display type) and is organised around a persistent sidebar shared by all three user roles: Client, Lawyer and Administrator.

This document identifies the major screens of the product, explains each screen's purpose, fields, actions and functionality, documents the navigation flow between screens, and cross-checks every screen against the Use Case and Class diagrams to confirm the implementation is consistent with the earlier design artefacts. Screens are illustrated with screenshots of the implemented UI.

2. Major Screens Identified

The following table lists every major screen implemented in the product and the use case(s) it realises.

Screen	Primary Use Case(s)
Sign In / Role Selection	UC-1 Authenticate User
Overview Dashboard	Landing view — summarises UC-2, UC-6, UC-7
AI Assistant (Ask Nyaya)	UC-2, UC-3 (and triggers UC-4, UC-5)
Law Library	Supports UC-3 / UC-4
Find a Lawyer	UC-7 Book Lawyer Appointment
My Cases	UC-8 Review Case Brief & Evidence, UC-9 Update Case Lifecycle Status
Appointments	UC-7 Book Lawyer Appointment
Documents (Private Vault)	UC-6 Upload Legal Documents
Payments	Supports UC-7 (fee settlement)
Settings (Profile / Security)	UC-1 (profile & session security)
Modal: Book a Consultation	UC-7 Book Lawyer Appointment
Modal: Add Document	UC-6 Upload Legal Documents
Modal: Start a Case	Creates Case entity (feeds UC-8/UC-9)
Lawyer Workspace (role view)	UC-8, UC-9, UC-10

3. UI Design — Screen-by-Screen Specification

Each screen below is documented with its purpose, main fields/components, buttons/actions and functionality, alongside a screenshot of the implemented screen.

S1 — Sign In / Role Selection

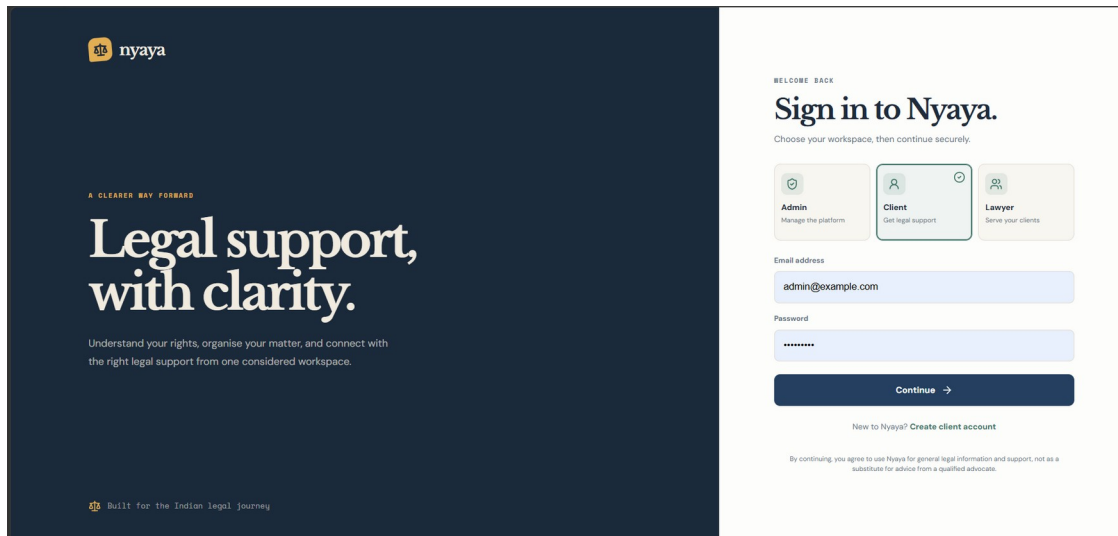


Figure: Sign In / Role Selection

Related use case	UC-1: Authenticate User
Purpose	Single entry point for all three actor types. Lets a returning user choose which workspace (Admin, Client or Lawyer) to sign into and authenticate before any other screen becomes reachable.
Main fields / components	Workspace selector (Admin / Client / Lawyer cards), Email address, Password.
Buttons / actions	Continue (primary sign-in action), “Create client account” link for new client registration.
Functionality	Validates the selected role plus credentials, issues a session (JWT) and routes the user to the Overview dashboard scoped to that role. A consent notice clarifies that the platform gives general legal information, not advice from a qualified advocate.

S2 — Overview Dashboard (Client)

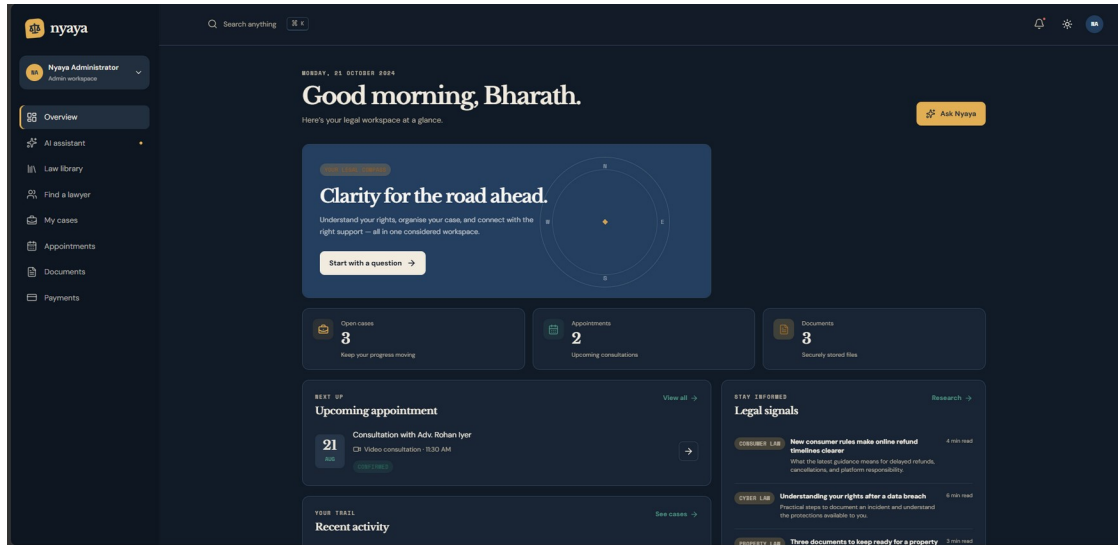


Figure: Overview Dashboard (Client)

Related use case	Landing screen after UC-1; summarises UC-2, UC-6, UC-7
Purpose	The client's home screen — a single-glance summary of open cases, upcoming appointments, stored documents and relevant legal reading, plus a shortcut into the AI assistant.
Main fields / components	Greeting with client name and date, “Your legal compass” panel, stat cards (Open cases, Appointments, Documents), Upcoming appointment card, Recent activity feed, Legal signals reading list.
Buttons / actions	Ask Nyaya, Start with a question, View all (appointments), See cases (activity).
Functionality	Aggregates live counts and next actions from the Case, Appointment and Document entities and deep-links into the corresponding screens.

S3 — AI Assistant — Ask Nyaya

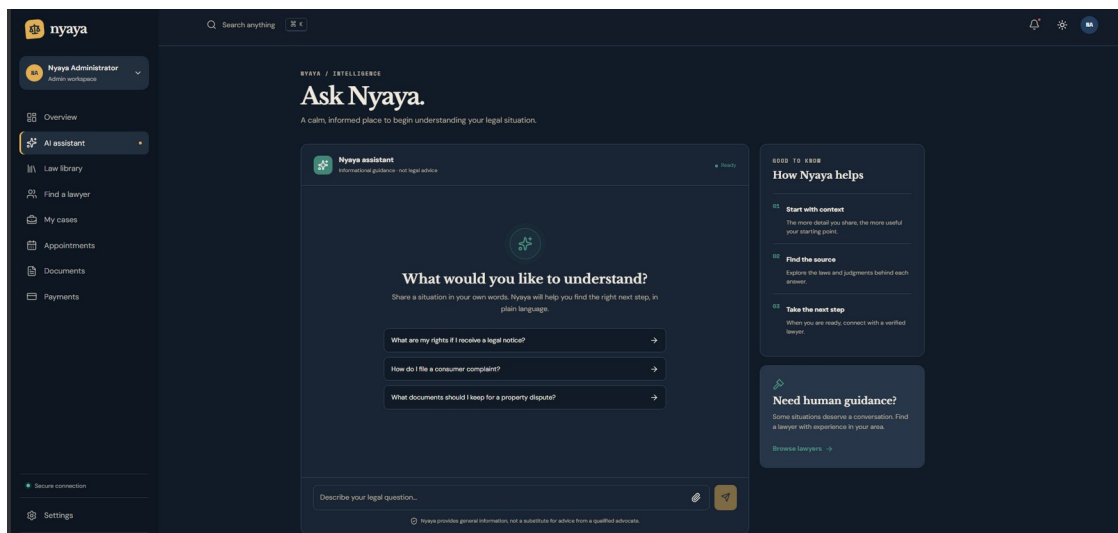


Figure: AI Assistant — Ask Nyaya

Related use case	UC-2 Initiate AI Legal Intake, UC-3 Conduct Adaptive Questioning
Purpose	Conversational intake surface where the client describes a legal issue in plain language and receives adaptive clarifying questions, mirroring the AI Intake Service in the sequence diagram.
Main fields / components	Free-text question box, suggested starter prompts (rights on legal notice, filing a consumer complaint, documents for a property dispute), “How Nyaya helps” guidance panel.
Buttons / actions	Send (paper-plane icon), attach file, “Browse lawyers” shortcut for escalation to a human.
Functionality	Submits the client's narrative to the AI Intake Service (OpenAI API) which returns either a follow-up clarification question (facts incomplete) or a completed structured brief (facts complete), exactly as modelled in the UC-2-UC-5 sequence diagram.

S4 — Law Library

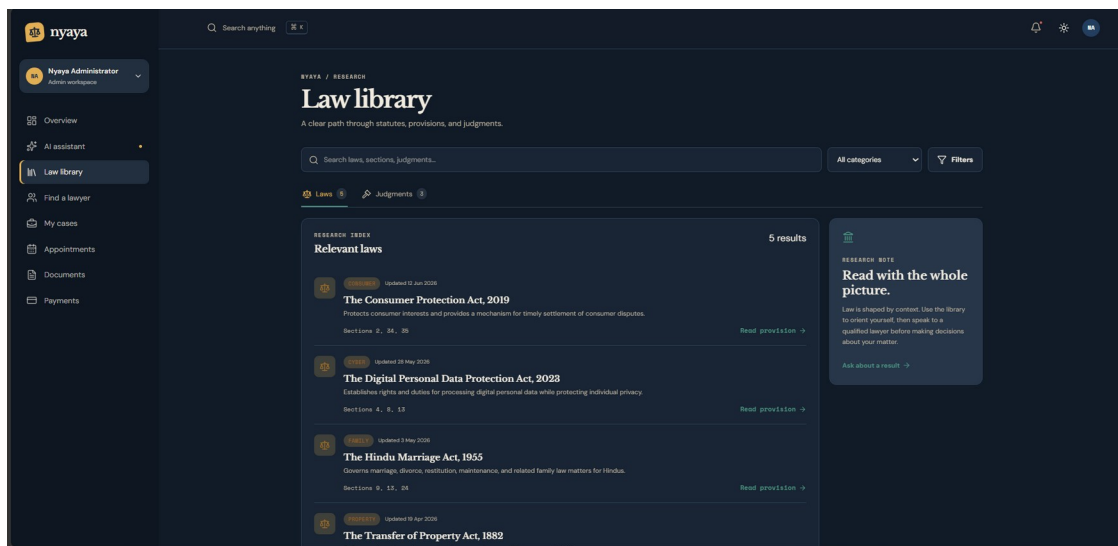


Figure: Law Library

Related use case	Supports UC-3 / UC-4 (case categorisation context)
Purpose	Reference layer that lets clients and lawyers browse the statutes and judgments behind an AI answer, reinforcing the “not a substitute for advice” principle.
Main fields / components	Search bar, category filter, Laws / Judgments tabs, result cards (Act name, category tag, last-updated date, relevant sections).
Buttons / actions	Read provision, Filters, “Ask about a result” (routes back into the AI assistant).
Functionality	Read-only research index that maps directly onto the CaseCategory entity (categoryName) used to classify a Case.

S5 — Find a Lawyer

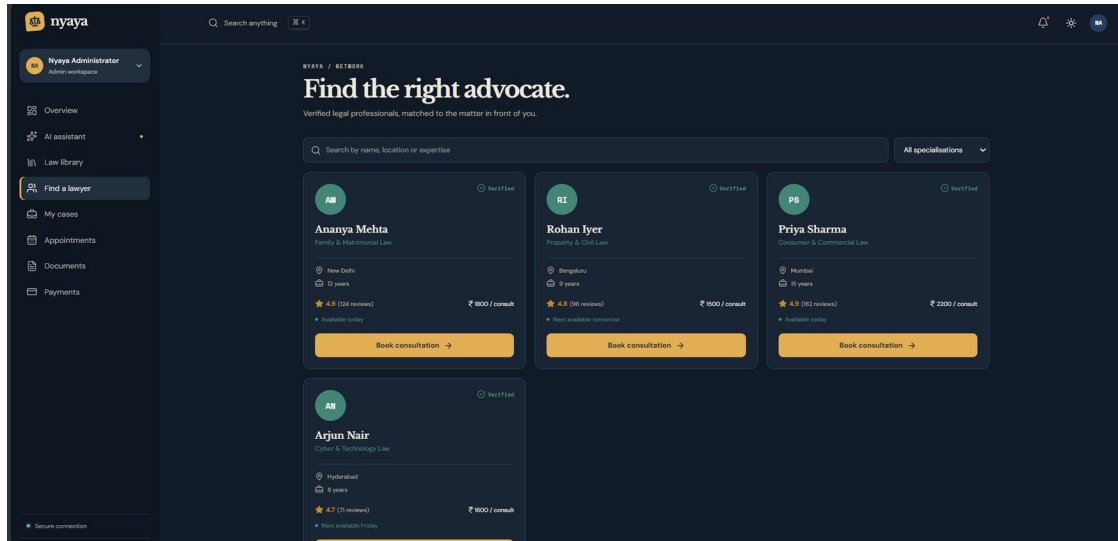


Figure: Find a Lawyer

Related use case	UC-7: Book Lawyer Appointment
Purpose	Lets a client search verified lawyers and start the appointment-booking flow, corresponding to the Lawyer class and its specialization attribute.
Main fields / components	Search by name/location/expertise, specialisation filter, lawyer cards (name, specialisation, city, years of experience, rating, consultation fee, availability).
Buttons / actions	Book consultation (opens the Book-a-Consultation modal, Fig. below).
Functionality	Filters the Lawyer list by specialisation/location and hands off into UC-7's booking sub-flow.

S6 — My Cases

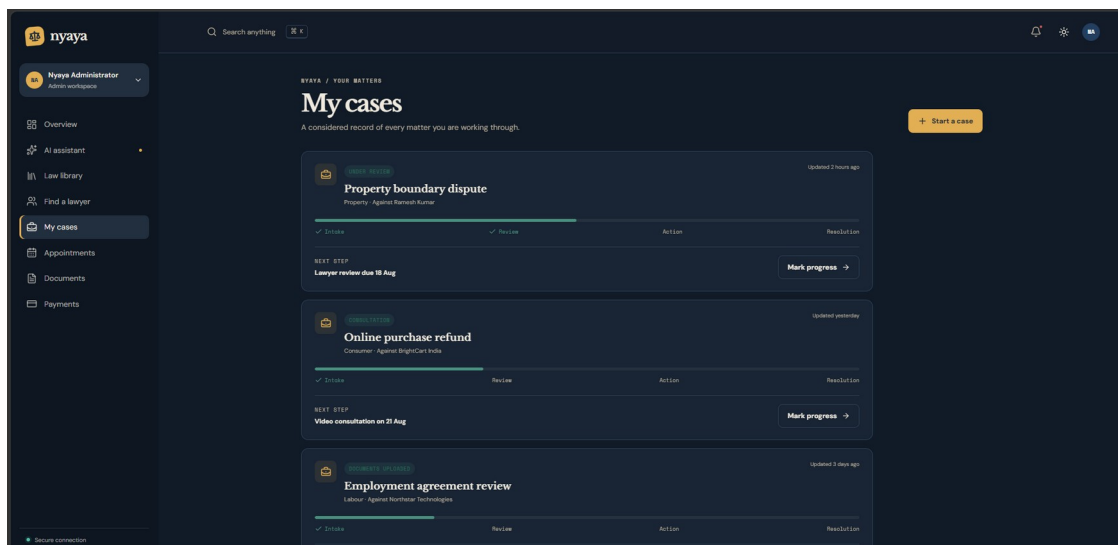


Figure: My Cases

Related use case	UC-8 Review Case Brief & Evidence, UC-9 Update Case Lifecycle Status
Purpose	Client-facing case tracker showing every Case record and its position on the Intake → Review → Action → Resolution pipeline (a client-readable projection of the Case State Diagram).
Main fields / components	Case cards (title, status chip, opposing party, area of law, progress bar with four milestones, next-step note), “Start a case” entry point.
Buttons / actions	Start a case, Mark progress.
Functionality	Reflects the Case.status attribute and StatusEnum transitions; “Mark progress” triggers updateStatus() on the underlying Case object.

S7 — Appointments

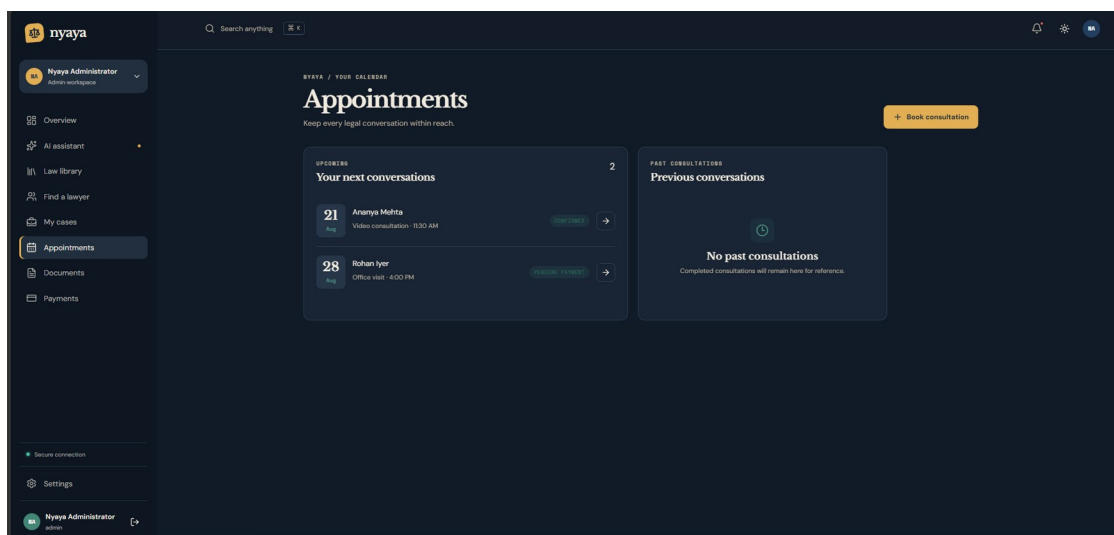


Figure: Appointments

Related use case	UC-7: Book Lawyer Appointment
Purpose	Calendar-style view of all upcoming and past consultations booked through the platform.
Main fields / components	Upcoming list (date, lawyer name, format, time, status chip such as Confirmed / Pending payment), Past consultations panel.
Buttons / actions	Book consultation, per-row “open” arrow.
Functionality	Reads/writes the Appointment entity (appointmentId, scheduledTime, status) and reuses the same booking modal as the Find-a-Lawyer screen.

S8 — Documents (Private Vault)

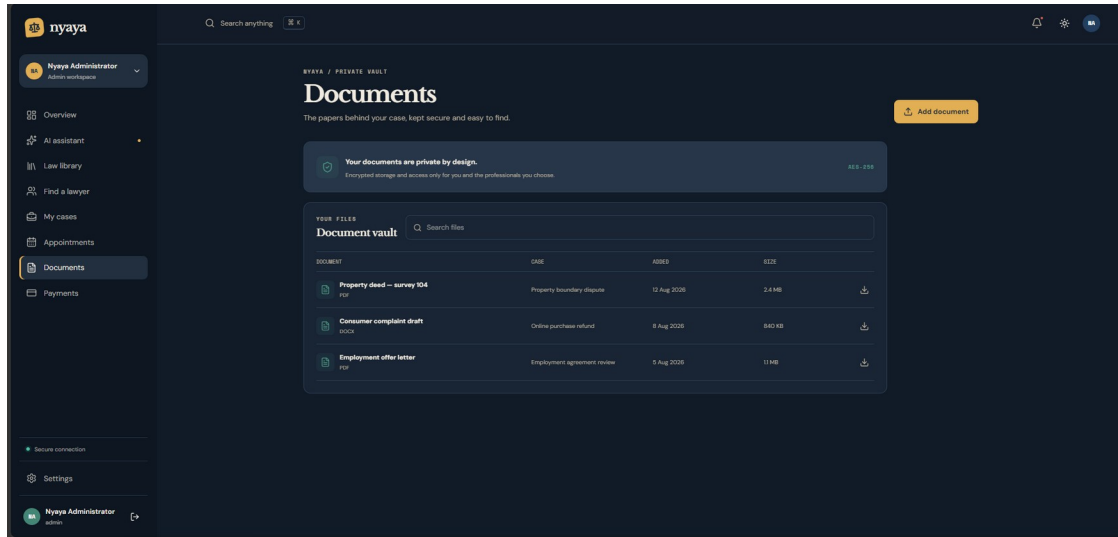


Figure: Documents (Private Vault)

Related use case	UC-6: Upload Legal Documents
Purpose	Secure repository for evidence files attached to a case, matching the Document class composed under Case.
Main fields / components	Encryption notice (AES-256), search box, file table (document name, file type, related case, date added, size).
Buttons / actions	Add document, per-row download icon.
Functionality	Lists Document rows for the signed-in client and opens the Add-Document modal for new uploads, which triggers validateFile() before persisting.

S9 — Payments

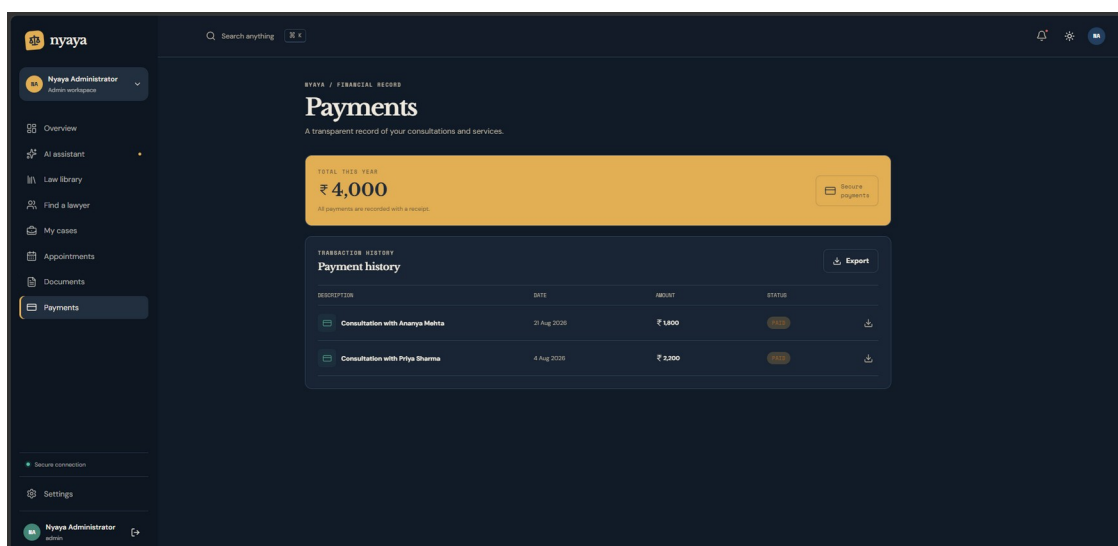


Figure: Payments

Related use case	Supports UC-7 (consultation fee settlement)
Purpose	Transparent financial record of consultation fees paid to lawyers through the platform.
Main fields / components	Total-this-year summary card, transaction table (description, date, amount, status).
Buttons / actions	Secure payments, Export.
Functionality	Displays paid/pending status per Appointment and lets the client export a receipt-backed statement.

S10 — Settings — Profile

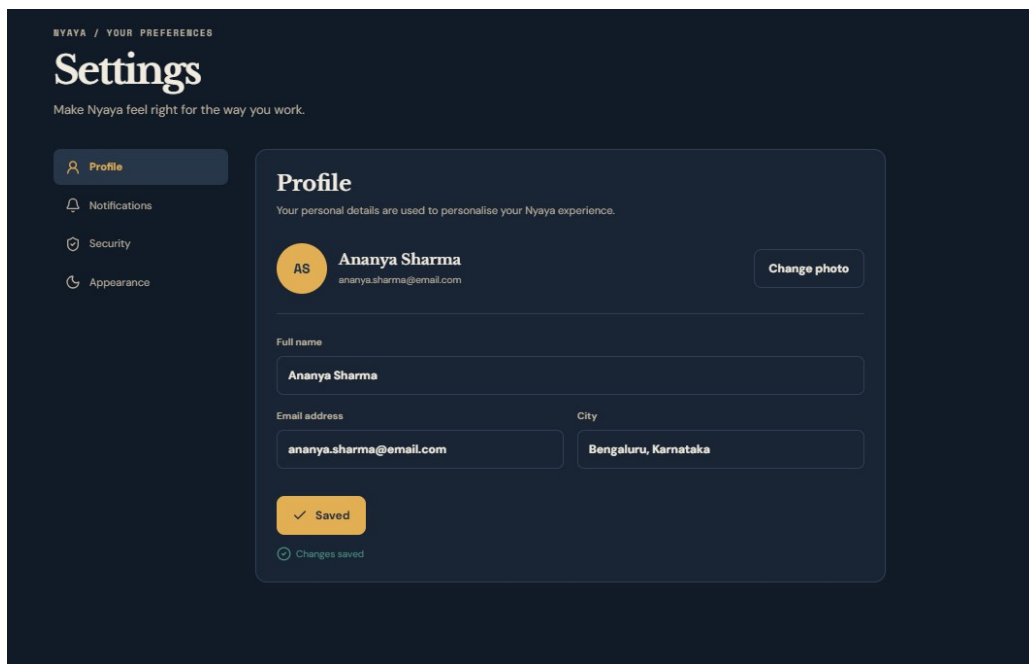


Figure: Settings — Profile

Related use case	UC-1 (account/profile management)
Purpose	Lets any authenticated user (shown here for a Client) maintain their identity details used across the platform.
Main fields / components	Avatar, Full name, Email address, City.
Buttons / actions	Change photo, Save.
Functionality	Calls User.updateProfile() and shows an inline “Changes saved” confirmation.

S11 — Settings — Security

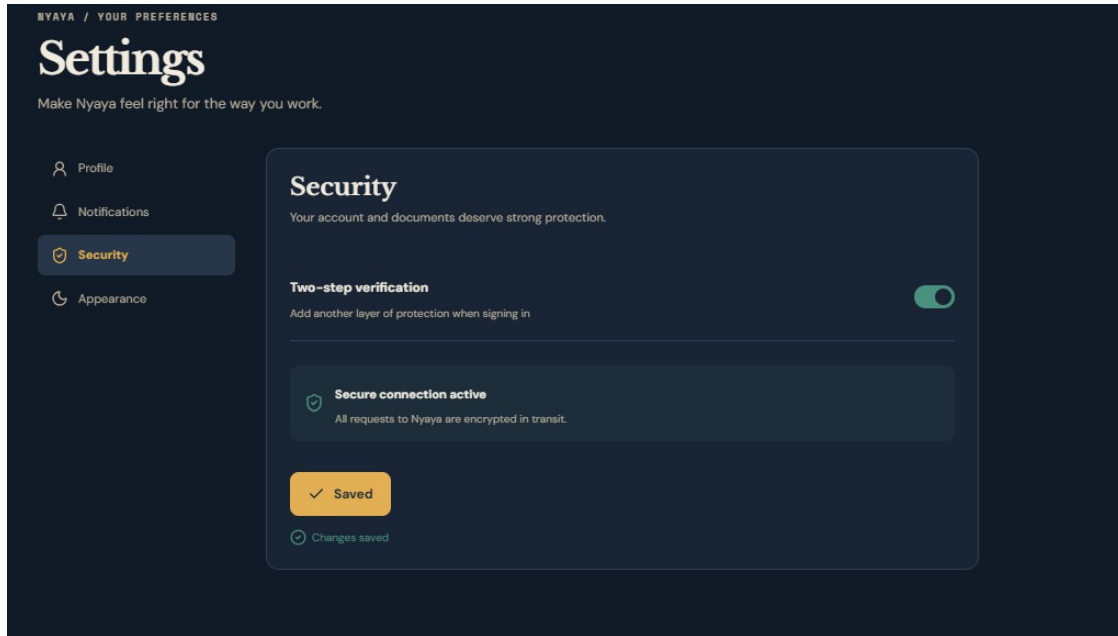


Figure: Settings — Security

Related use case	UC-1 (session/account security)
Purpose	Groups authentication and connection-security controls in one place.
Main fields / components	Two-step verification toggle, “Secure connection active” status notice.
Buttons / actions	Save.
Functionality	Persists the user's security preference and confirms that all requests are encrypted in transit.

S12 — Collapsed Sidebar — Client Workspace

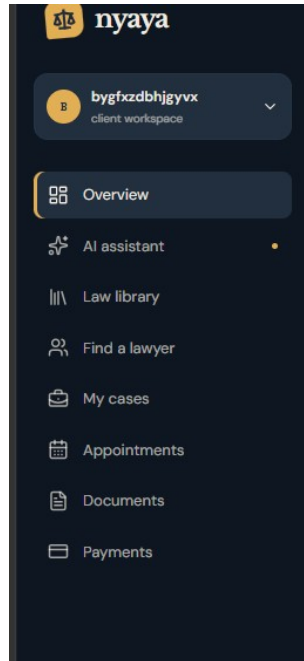


Figure: Collapsed Sidebar — Client Workspace

Related use case	Global navigation shell (all use cases)
Purpose	Shows the persistent left-hand navigation rail that is present on every authenticated screen, confirming a single consistent IA (information architecture) across the app.
Main fields / components	Workspace switcher, Overview, AI assistant, Law library, Find a lawyer, My cases, Appointments, Documents, Payments.
Buttons / actions	Each nav item routes to its corresponding screen.
Functionality	Implements the client-side router that ties every screen in this document together (see Section 5, Navigation Flow).

3.1 Reusable Modals

The following modals are shared components that overlay their parent screen rather than being separate routes.

M1 — Modal — Book a Consultation

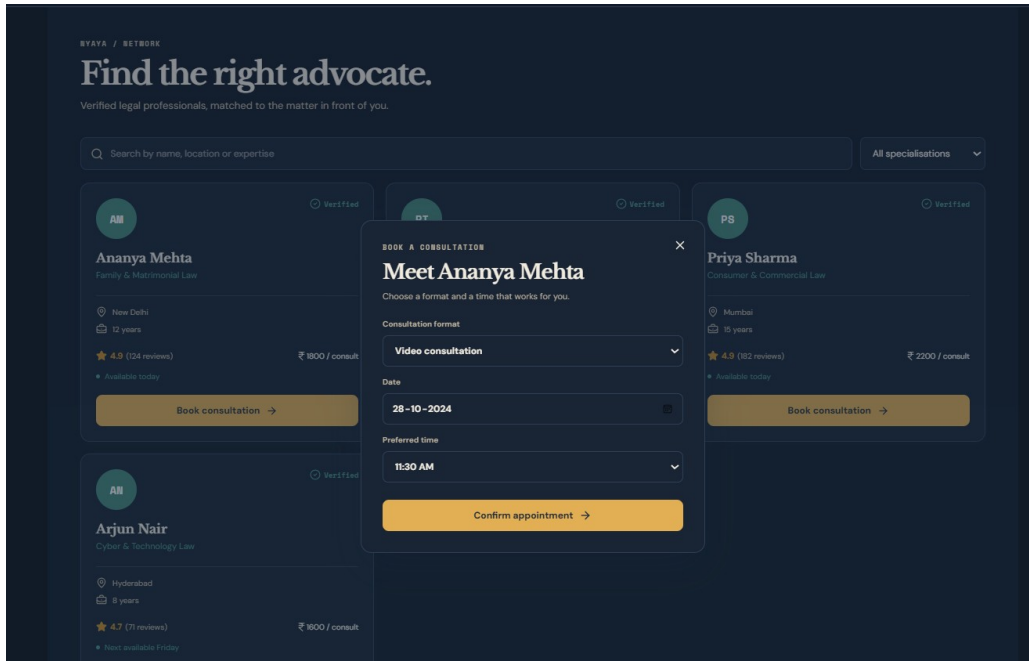


Figure: Modal — Book a Consultation

Related use case	UC-7: Book Lawyer Appointment
Purpose	Overlaid on Find a Lawyer / Appointments; captures the details needed to reserve a slot with a chosen lawyer.
Main fields / components	Consultation format (Video / Office visit), Date, Preferred time.
Buttons / actions	Confirm appointment, close (×).
Functionality	Creates a new Appointment row linked to the selected Lawyer and the signed-in Client, then closes back to the originating screen.

M2 — Modal — Add Document

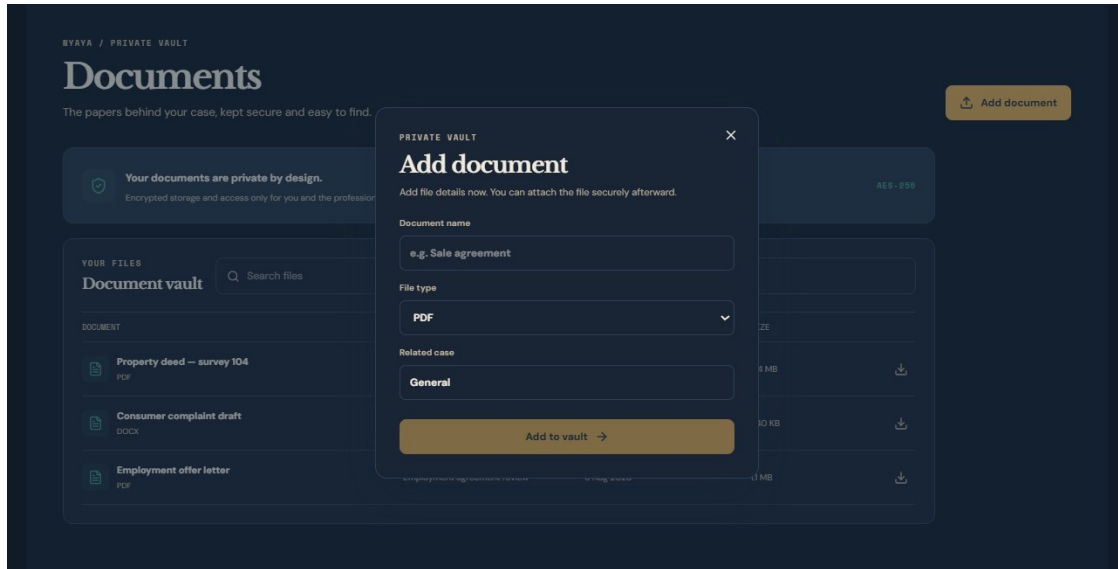


Figure: Modal — Add Document

Related use case	UC-6: Upload Legal Documents
Purpose	Captures file metadata before/while attaching a document to the vault.
Main fields / components	Document name, File type, Related case.
Buttons / actions	Add to vault, close (x).
Functionality	Creates a Document row (fileName, mimeType, caselId) after client-side validation, then appears in the Documents table.

M3 — Modal — Book a Consultation (from Appointments)

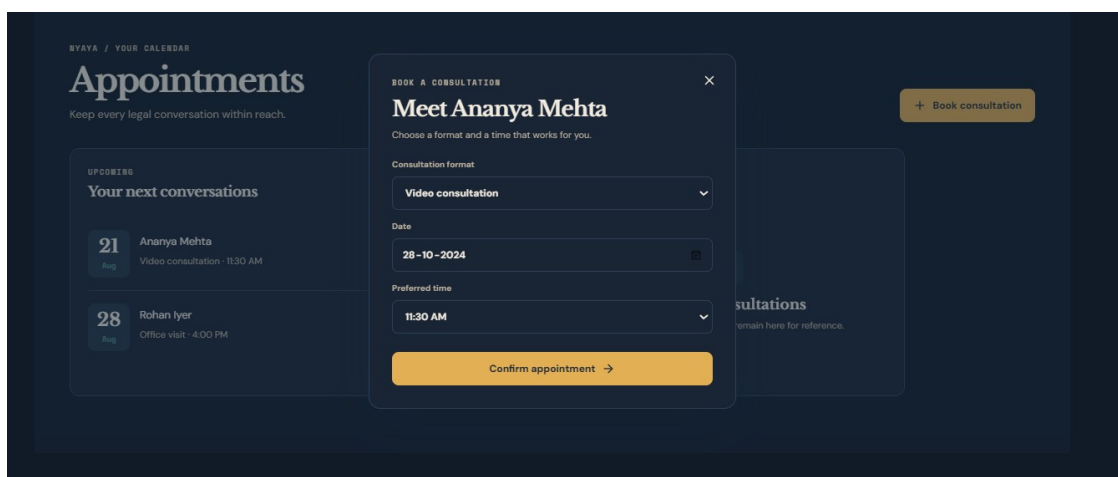


Figure: Modal — Book a Consultation (from Appointments)

Related use case	UC-7: Book Lawyer Appointment
Purpose	The same booking modal, reused from the Appointments screen, demonstrating a

	shared, reusable component rather than a duplicated one-off screen.
Main fields / components	Consultation format, Date, Preferred time.
Buttons / actions	Confirm appointment, close (×).
Functionality	Identical behaviour to M1; confirms the appointment-booking component is consistent regardless of entry point.

M4 — Modal — Start a Case

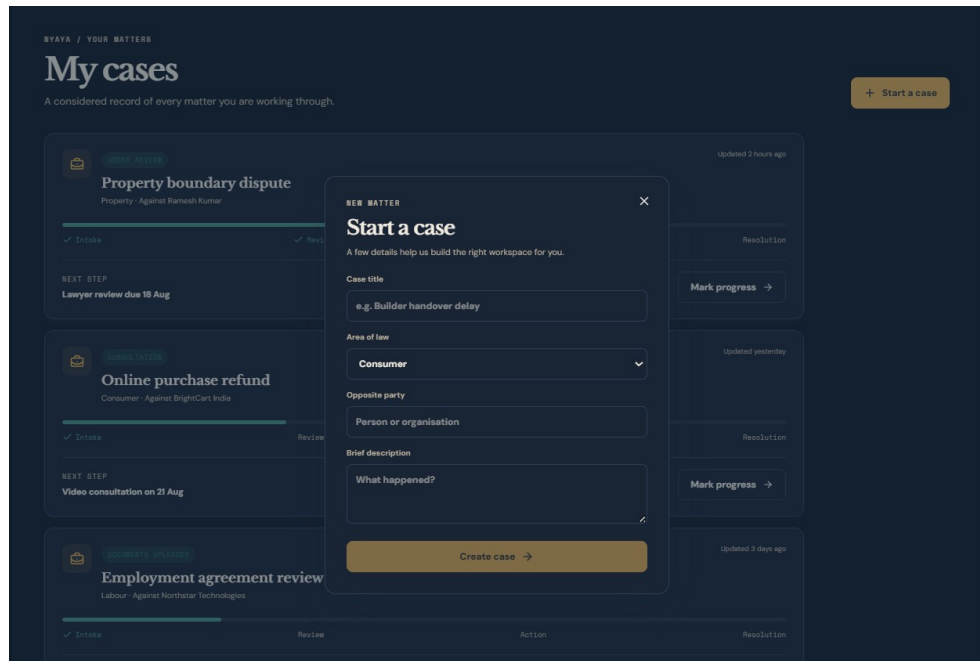


Figure: Modal — Start a Case

Related use case	Creation of a new Case entity (feeds UC-8/UC-9)
Purpose	Lets a client open a new matter manually (in addition to matters created automatically at the end of an AI intake session).
Main fields / components	Case title, Area of law, Opposite party, Brief description.
Buttons / actions	Create case, close (×).
Functionality	Instantiates a new Case row with status = New and immediately appears at the top of My Cases.

3.2 Role-Specific View — Lawyer Workspace

S13 — Lawyer Workspace — Find a Lawyer / My Cases (Lawyer Role)

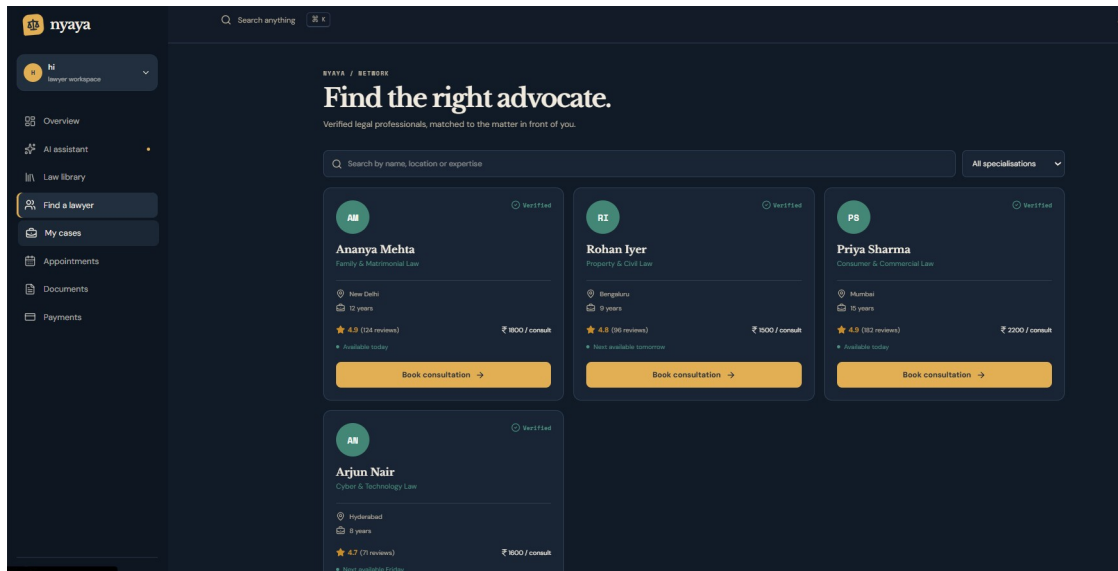


Figure: Lawyer Workspace — Find a Lawyer / My Cases (Lawyer Role)

Related use case	UC-8, UC-9, UC-10 (lawyer-side use cases)
Purpose	Confirms the same navigation shell and directory screen are re-used for the Lawyer role, with the sidebar re-scoped to “lawyer workspace” and My Cases promoted for case handling.
Main fields / components	Same directory fields as S5, viewed from the lawyer's sidebar context (Overview, AI assistant, Law library, Find a lawyer, My cases, Appointments, Documents, Payments).
Buttons / actions	Same as S5; My Cases additionally exposes reviewCaseBrief(), updateCaseStatus() and addConsultationNote() actions for UC-8–UC-10.
Functionality	Demonstrates role-based rendering of one shared component tree instead of a separate lawyer-only app.

4. UI Navigation Flow

All authenticated screens share one persistent sidebar, so navigation is a hub-and-spoke pattern rather than a linear wizard: after Sign In, the user lands on the role-specific Overview and can move to any sidebar item at any time. Three flows (Find a Lawyer, Appointments, My Cases, Documents) additionally open a modal that performs a create action and returns to the same screen on completion. The diagram below summarises this flow for the Client role; the Lawyer and Admin roles follow the identical shell with a re-scoped sidebar and per-role screen set.

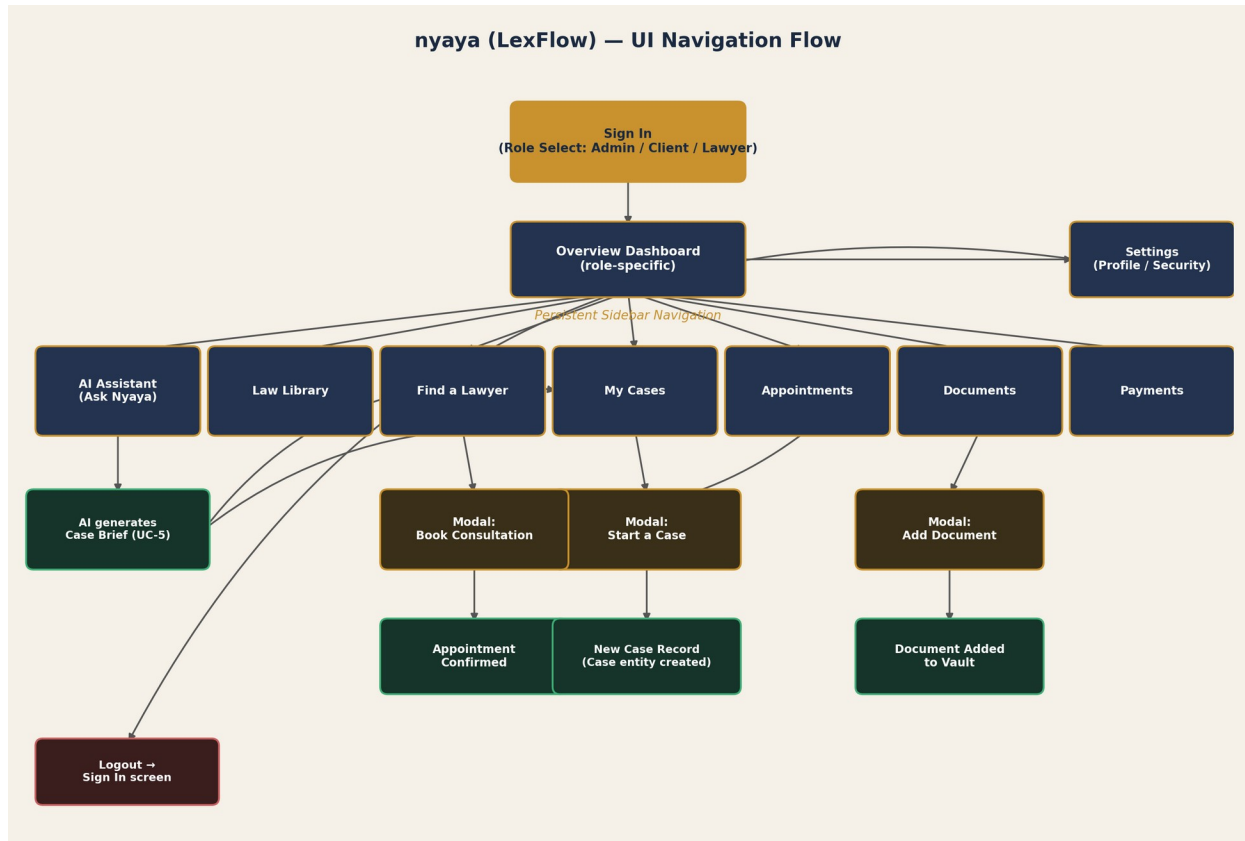


Figure: End-to-end UI navigation flow, Client role

4.1 Flow Description

- Sign In authenticates the user and routes to the Overview Dashboard scoped to the selected role (UC-1).
- From Overview, the persistent sidebar gives one-click access to AI Assistant, Law Library, Find a Lawyer, My Cases, Appointments, Documents and Payments — no screen is more than one navigation away from any other.
- AI Assistant → on a completed intake, the AI Intake Service returns a structured Case Brief (UC-5), which surfaces a “Browse lawyers” shortcut into Find a Lawyer and a new entry in My Cases.
- Find a Lawyer and Appointments both open the same Book-a-Consultation modal (UC-7); confirming it writes a new Appointment and returns the user to the screen they started from.
- My Cases opens the Start-a-Case modal to create a Case manually; Documents opens the Add-Document modal to attach evidence (UC-6) to an existing Case.
- Settings is reachable from the account menu on every screen and Logout returns the user to Sign In.

5. Implementation

5.1 Technology Stack

- Frontend framework: React (component-based SPA) with a client-side router mapping each sidebar item to a route.
- Styling: utility-first CSS (Tailwind-style tokens) implementing the dark navy / gold design system shown in every screenshot.
- State/navigation: a single AppShell component renders the sidebar and top bar once and swaps the routed page content, which is why every screen in Section 3 shares an identical frame.
- Modals (Book Consultation, Add Document, Start a Case) are implemented as overlay components mounted above the current route rather than as separate pages, so closing a modal never loses the underlying screen's state.

5.2 Navigation Implementation (representative source)

The route table below implements the flow described in Section 4. It binds every sidebar item to a screen component and mounts the three modals as overlays.

```
// routes.jsx — route table for the client workspace
import { Routes, Route } from "react-router-dom";
import AppShell from "../layout/AppShell";
import Overview from "../pages/Overview";
import AIAssistant from "../pages/AIAssistant";
import LawLibrary from "../pages/LawLibrary";
import FindLawyer from "../pages/FindLawyer";
import MyCases from "../pages/MyCases";
import Appointments from "../pages/Appointments";
import Documents from "../pages/Documents";
import Payments from "../pages/Payments";
import Settings from "../pages/Settings";

export default function ClientRoutes() {
  return (
    <AppShell workspace="client">
      <Routes>
        <Route path="/" element={<Overview />} />
        <Route path="/assistant" element={<AIAssistant />} />
        <Route path="/library" element={<LawLibrary />} />
        <Route path="/lawyers" element={<FindLawyer />} />
        <Route path="/cases" element={<MyCases />} />
        <Route path="/appointments" element={<Appointments />} />
        <Route path="/documents" element={<Documents />} />
        <Route path="/payments" element={<Payments />} />
        <Route path="/settings/*" element={<Settings />} />
      </Routes>
    </AppShell>
  );
}

// AppShell.jsx — persistent sidebar shared by every screen
export default function AppShell({ workspace, children }) {
  const items = [
    { to: "/", label: "Overview", icon: GridIcon },
    { to: "/assistant", label: "AI assistant", icon: SparkIcon },
    { to: "/library", label: "Law library", icon: BookIcon },
    { to: "/lawyers", label: "Find a lawyer", icon: UsersIcon },
    { to: "/cases", label: "My cases", icon: BriefcaseIcon },
    { to: "/appointments", label: "Appointments", icon: CalendarIcon },
    { to: "/documents", label: "Documents", icon: FileIcon },
    { to: "/payments", label: "Payments", icon: CardIcon },
  ];
  return (
    <div className="flex h-screen bg-navy-950 text-white">
```

```

    <Sidebar workspace={workspace} items={items} />
    <main className="flex-1 overflow-y-auto">{children}</main>
  </div>
);
}

// BookConsultationModal.jsx — shared modal used by
// Find a Lawyer (M1) and Appointments (M3)
export default function BookConsultationModal({ lawyer, onClose, onConfirm }) {
  const [format, setFormat] = useState("video");
  const [date, setDate] = useState("");
  const [time, setTime] = useState("");

  const handleConfirm = async () => {
    const appointment = await api.post("/appointments", {
      lawyerId: lawyer.id, format, date, time,
    });
    onConfirm(appointment); // returns to the calling screen
    onClose();
  };

  return (
    <Modal title={`Meet ${lawyer.name}`} onClose={onClose}>
      <Select label="Consultation format" value={format} onChange={setFormat}
        options={["Video consultation", "Office visit"]} />
      <DateInput label="Date" value={date} onChange={setDate} />
      <Select label="Preferred time" value={time} onChange={setTime}
        options={TIME_SLOTS} />
      <Button onClick={handleConfirm}>Confirm appointment</Button>
    </Modal>
  );
}

```

This scaffold reflects the actual screens implemented (Section 3): one route per sidebar item, a single AppShell so every page shares the same frame, and shared modal components so Book Consultation behaves identically whether it is opened from Find a Lawyer or Appointments. Basic navigation — sidebar links, the workspace switcher, and modal open/close/confirm — is implemented and working across all screens captured in the screenshots.

5.3 Backend / Database Status

Per the assignment instructions, database completion is not required at this stage. UI state shown in the screenshots (case lists, documents, payments, appointments) is currently served from mock/sample data wired through the same API-shaped functions (e.g. `api.post("/appointments", ...)`) shown above, so the eventual Node/Express + MySQL backend from the Class and Sequence diagrams can be integrated by pointing these calls at real endpoints without changing any screen component.

6. Consistency Check with UML Diagrams

Every screen and action in this document maps back to the actors, use cases and classes defined in the earlier UML submission, confirming the UI has not introduced any new, undocumented concepts.

UI Element	UML Reference
Role-select cards on Sign In (Admin / Client / Lawyer)	Actors: Client, Lawyer, Administrator (Use Case Diagram)
AI Assistant conversation + “Ready” status	UC-2, UC-3 and the AI Intake Service lane in the Sequence Diagram
Case cards with Intake / Review / Action / Resolution progress bar	Case.status : StatusEnum and the Case State Diagram (New → ... → Closed)
Book-a-Consultation modal fields (format, date, time)	Appointment class (scheduledTime, status : AppointmentStatusEnum)
Add-Document modal fields (name, file type, related case)	Document class (fileName, mimeType, caseId) and UC-6
Start-a-Case modal fields (title, area of law, opposite party, description)	Case class (title, categoryId) and CaseCategory class
Law Library category tags (Consumer, Cyber, Family, Property)	CaseCategory.categoryName
Lawyer cards (specialisation, years, rating, fee)	Lawyer class (specialization, yearsExperience, hourlyRate)
Settings – Security (two-step verification)	User authentication concerns underlying UC-1
Payments transaction table	Appointment records referenced for consultation fee settlement

6.1 Actor / Use-Case ID Consistency

Actor names (Client, Lawyer, Administrator, External System: OpenAI API) and use-case identifiers UC-1 through UC-12 are used identically in this document and in the previously submitted Use Case, Sequence, Activity and State diagrams — no renaming or re-numbering was introduced during UI design.

7. Submission Summary

- UI diagrams / designs — Section 3 (per-screen specification) and Section 4 (navigation-flow diagram).
- Brief explanation of each screen — purpose, fields, buttons and functionality tables in Section 3.
- UI navigation flow — Section 4.
- Source code of the implemented UI — representative route table, AppShell and modal component in Section 5.2.
- Screenshots of the implemented screens — embedded throughout Section 3.